IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

SOFTWARE SYSTEMS

Monday 15 May 14:30

Time allowed: 2 hours

There are THREE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S. Baig
Second Marker(s) : J.V. Pitt

© Imperial College London

Software Systems

Marks: 60

Question 1

(a) Draw an Entity Relationship diagram for the following description of a bank's software system. Show the entities, relationships, cardinalities of relationships, participation types of relationships, attributes, and primary keys.

The bank has several customers, branches, and accounts. Each branch has a unique branch code and a branch name. Each customer has a unique id, name, address, and the total amount across all their accounts. Each account has a unique account number (unique across all branches of the bank), the amount of money currently in the account and an account type. The possible account types are personal, business, joint, salary, savings and current. The same account may be of more than one types, e.g., a personal account may be a joint savings account, etc. An account belongs to a specific branch.

An account may be owned by one or more customers (in case of joint accounts). A customer may own one or more accounts. There cannot be a customer who does not own an account anywhere. Similarly, there cannot be an unowned account.

A customer can deposit money into an account and withdraw money from it. Each deposit and withdrawal has an associated amount (the amount deposited or withdrawn) and a timestamp. A customer can make any number of deposits and withdrawals. Similarly, an account may be deposited into and withdrawn from any number of times.

A customer may transfer money to another customer. A transfer has an associated sender and receiver account numbers, the amount of money transferred, and a timestamp. It is possible for a customer to never withdraw money or participate in a transfer. However, each customer must make an initial deposit into each of their accounts. Similarly, an account must be deposited into at least once, however, it is possible that money is never withdrawn from it.

A branch can give a loan to a customer. When giving the loan, another customer must act as the guarantor. A loan has an associated amount, interest rate, borrower's account number, and timestamp. A customer may borrow no more than one loan at a time. Similarly, a customer may not act as the guarantor in more than one loans at a time. A branch can hand out any number of loans.

[10]

 © 2023 Imperial College London

(b) Convert the Entity Relationship model from (a) into a relational schema. Highlight the primary key of each relation in the schema.

[5]

(c) The following relational schema represents a miniature course enrolment system.

```
Student(student_id, student_name, average_grade)
Enrolment(student_id, course_id, date)
Course(course_id, course_title, department, credit_hours)
Teaches(lecturer_id, course_id)
Lecturer(lecturer_id, lecturer_name, department)
```

Write SQL queries to perform the following tasks:

(i) List the ids and names of students who have only enrolled in courses offered by the department of 'Computer Science.' (You may assume that every student has enrolled in at least one course).

(ii) List the ids and names of students who are being taught by three or more different lecturers.

Note: You may use NATURAL JOIN to join tables on columns with identical names, e.g., *Student* NATURAL JOIN *Enrolment* will join the two tables on student_id.

[5 + 5]

Question 2

(a) Answer the following questions related to application and transport layer protocols:

i. Give two different examples of application layer protocols that rely on caching to improve the user perceived performance of the network application. Indicate why caching is used, where the caches are located, and how they work.

ii. Indicate if TCP or UDP would be a better fit for the service requirements of the following applications and briefly explain why. State any assumptions that you make for each application:

– Online multiplayer first-person shooting game
– Text based instant messaging app
– Clock synchronization between computers on the internet

[3 + 3]

(b) Suppose a browser is fetching a webpage. The base HTML file references 13 very small objects on the same server. Assuming: an RTT of 3ms, each object small enough to fit a

single TCP segment, no transmission delays, how much time does it take to fetch the complete page, when using:

(i) Non-persistent HTTP with no parallel TCP connections?
(ii) Persistent HTTP with no parallel TCP connections?
(iii) Non-persistent HTTP with 3 parallel connections at max. Note: the connections can only be created in serial.
(iv) Persistent HTPP with no parallel TCP connections, just like in (ii). However, this time we also wish to account for the DNS query. Assume that we need to visit 4 servers (including the local DNS) before the IP address is retrieved. Assume DNS querying to be iterative and the transport layer protocol of DNS to be UDP. All RTTs are 3ms.

[2+2+3+2]

(c) Host X needs to send a message consisting of 9 packets to host Y. The two hosts are using a sliding window Go-Back-N strategy with a window size of 3. All packets are available for transmission immediately (so as many packets as the window size allows can be sent at once). Assume that every 6th packet that X transmits gets lost. But no ACKs from Y ever get lost. Also assume that the medium delivers packets in order. What is the number of packets that X will need to transmit in order to send the entire message to B?

[5]

Question 3

(a) This question is about Network Address Translation (NAT)

(i) Suppose a router performing NAT with public address 205.10.221.50 is operating on behalf of the nodes in your home, which are part of the internal private network 192.168.4.0/24. When a client with address 192.168.4.2 sends a TCP packet to an outside server at 201.20.4.7 (as shown in the table below), how does NAT modify the packet?

Original packet (before NAT)

	Source	Destination
IP	192.168.4.2	205.10.221.50
TCP Port	5555	20

After NAT

	Source	Destination
IP	?	?
TCP Port	?	?

(ii) What information (mapping) does the router need to keep internally in order to properly translate and forward the server's response?

[3+3]

(b) This part relates to subnetting and IP addressing. Suppose an ISP owns the block of addresses of the form 145.126.252.0/22. The ISP wishes to create three subnets from this block: Subnet 1, 2 and 3. Subnet 1 needs a minimum of 350 interfaces, Subnet 2 needs a minimum of 105 interfaces, and Subnet 3 needs a minimum of 221 interfaces. Provide three network addresses (of the form a.b.c.d/x) for the three subnets that satisfy these constraints. Make sure the address spaces don't overlap.

[4]

(c) The following figure shows a network of hosts, routers and switches.

(i) How many subnets does the network have?

Nodes H2 to H4 connected to the network before H1, all using DHCP. Finally, H1 connected to the network using DHCP.

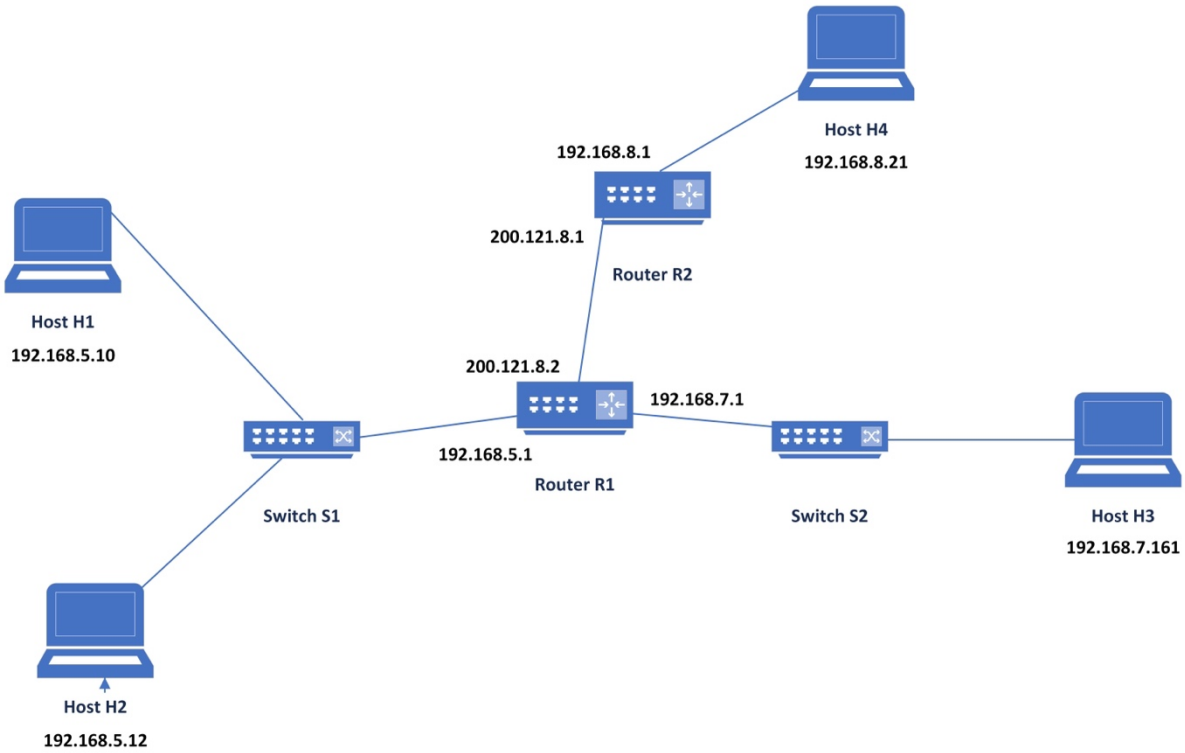
(ii) Which node (or nodes) would have received the DHCP request from H1? Which node (or nodes) would have responded to this request?

Now H1 needs to send an IP datagram to H3 (H1 knows the IP address of H3). However, H1's ARP table is empty, hence it needs to send out an ARP request. The ARP tables at all other nodes are fully populated and up to date.

(iii) For which IP address does H1 send the ARP query to get its corresponding MAC address?

(iv) Which nodes receive H1's ARP request? Which nodes respond to H1's ARP request?

(v) How many more ARP requests need to be made before the packet reaches its destination H3?


Host H4
192.168.8.21
192.168.8.1
200.121.8.1
Router R2
Host H1
192.168.5.10
200.121.8.2
192.168.7.1
192.168.5.1
Router R1
Switch S1
Switch S2
Host H3
192.168.7.161
Host H2
192.168.5.12

[1+1+1+1+1]